

Nikita Kiselev

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Check my Resume (one-page version)

Curriculum Vitæ

SUMMARY

Senior ML Research Engineer at Kandinsky Lab. Develops SFT and RLHF for Kandinsky 6.0 text-to-audio-video and image-to-audio-video. Previously built post-training and alignment for Kandinsky 5.0 and 4.1 image and video models. Strong mathematical background in optimization and loss-landscape geometry. Lecturer in deep learning and generative models.

Professional Skills

1. **Generative AI:** diffusion models, text-to-audio-video, image-to-audio-video, text-to-video, image-to-video, text-to-image, image-to-image, alignment (SFT, RLHF), controllable generation (ControlNet), adapters (IP-Adapter, Face-Adapter), style transfer, instruction-based image editing, image quality assessment.
2. **Computer Vision:** medical images reconstruction and forecasting (fMRI, EEG).
3. **Optimization:** sample size determination, loss landscape, decentralized optimization.

WORK EXPERIENCE

Kandinsky Lab, Sber

Senior Research Engineer

Kandinsky 6.0 (text-to-audio-video, image-to-audio-video):

- Post-Training: SFT.
- Alignment: RLHF.

Moscow, Russia

Aug 2026 – Present

Kandinsky Lab, Sber

Research Engineer

Kandinsky 5.0 (text-to-image, image-to-image):

- Post-Training: SFT, Instruct SFT.
- Alignment: RLHF (DPO, DRAFT, GRPO, DGPO).
- Applications: Text Rendering, Interior Design, Stylization.

Kandinsky 4.1 (text-to-image, text-to-video, image-to-video):

- Performed text-to-image SFT via innovative LoRA-Soup technique, significantly improved image generation quality.
- Text-to-video LoRA fine-tuning for Russian culture code. Trained 70+ LoRA adapters on different characters, distilled them into text-to-video generation of Kandinsky 4.1.
- Image-to-video generation of Kandinsky 4.1. Developed a new architecture and trained the model maintaining structural dynamics.

Moscow, Russia

Jan 2025 – Jul 2026

Research Intern

Jun 2024 – Dec 2024

Kandinsky Image Editing:

- Interior Design with Kandinsky 3 ControlNet for Domclick. Fine-tuned ControlNet on the interiors domain, implemented regional prompting for arbitrary segmentation masks and utilized ControlNet image2image editing.
- Kandinsky 3 ControlNet Style (DepthPro) that outperforms ControlNet Union SDXL by 100% on ImageReward. Trained ControlNet over 300k steps on the 3M dataset, used DepthPro condition for the best image preserving and style editing.

Research Center for Artificial Intelligence, Innopolis University

Mathematician-programmer

- Research work in the field of optimization

Laboratory of Mathematical Methods of Optimization, MIPT

Technician

- Research work in the field of optimization
- Decentralized optimization with coupled constraints
- Network design problem

Innopolis, Russia

Sep 2024 – Jul 2025

Moscow, Russia

Oct 2023 – Apr 2024

EDUCATION

Moscow Institute of Physics and Technology

PhD in Artificial Intelligence and Machine Learning

Moscow, Russia

Sep 2026 – Present

- Thesis: Scaling of Parametric Machine Learning Models Based on the Local Geometry of the Optimization Surface
- Advisor: Andrey Grabovoy, PhD

Moscow Institute of Physics and Technology

MSc in Computer Science

Moscow, Russia

Sep 2024 – Aug 2026

- Thesis: Sample Size Determination for Parametric Models
- Advisor: Andrey Grabovoy, PhD
- GPA: 5/5

Moscow Institute of Physics and Technology

BSc in Applied Mathematics and Physics

Moscow, Russia

Sep 2020 – Aug 2024

- Thesis: Bayesian Sample Size Estimation
- Advisor: Andrey Grabovoy, PhD
- GPA: 4.88/5 (with honours)

PUBLICATIONS

Conference Proceedings.....

1. D. Yarmoshik, A. Rogozin, N. Kiselev, D. Dorin, A. Gasnikov, D. Kovalev
Decentralized Optimization with Coupled Constraints
ICLR 2025, A*
2. V. Meshkov, N. Kiselev, A. Grabovoy
ConvNets Landscape Convergence: Hessian-Based Analysis of Matricized Networks
2024 Ivannikov Ispras Open Conference (ISPRAS)

Journal Papers.....

1. N. Kiselev, V. Meshkov, A. Grabovoy
Robust Convergence of Loss Landscapes through Distributional Averaging
Proceedings of the Institute for System Programming of the RAS
2. D. Dorin, N. Kiselev, A. Grabovoy
Decoding visual information from neural signals: image reconstruction based on joint functional magnetic resonance imaging and electroencephalography analysis
Informatics and Applications
3. N. Kiselev, A. Grabovoy
Sample Size Determination: Posterior Distributions Proximity
Computational Management Science
4. N. Kiselev, A. Grabovoy
Sample Size Determination: Likelihood Bootstrapping
Computational Mathematics and Mathematical Physics
5. N. Kiselev, A. Grabovoy
Unraveling the Hessian: A Key to Smooth Convergence in Loss Function Landscapes
Doklady Mathematics
6. D. Dorin, N. Kiselev, A. Grabovoy, V. Strijov

Technical Reports.....

1. Kandinsky Lab, Sber
Kandinsky 6.0 Video: Foundation Models for Synchronized Video and Audio Generation
2. Kandinsky Lab, Sber
Kandinsky 5.0: A Family of Foundation Models for Image and Video Generation
Top-1 Daily & Weekly & Monthly Paper on Hugging Face (November 2025)

Preprints.....

1. E. Petrov, N. Kiselev, V. Meshkov, A. Grabovoy
Closing the Curvature Gap: Full Transformer Hessians
2. N. Kiselev, A. Grabovoy
Curvature-Aligned Probing for Local Loss-Landscape Stabilization

Conference Abstracts.....

1. N. Kiselev, A. Grabovoy
A mean-square criterion for loss-landscape convergence based on the leading eigenspace of the Hessian
Proceedings of the 68th MIPT All-Russian Scientific Conference
2. N. Kiselev, V. Meshkov, A. Grabovoy
Sufficient sample size and its connection to the loss landscape convergence
Proceedings of the 22nd All-Russian Conference with International Participation "Mathematical Methods of Pattern Recognition"
3. D. Dorin, N. Kiselev, A. Grabovoy
Decoding visual stimuli from multimodal brain signals
Proceedings of the 22nd All-Russian Conference with International Participation "Mathematical Methods of Pattern Recognition"
4. N. Kiselev, A. Grabovoy
Loss landscape convergence as an indicator of sufficient sample size
Proceedings of the 67th MIPT All-Russian Scientific Conference
5. N. Kiselev, A. Grabovoy
Determining a sufficient sample size from the posterior distribution of model parameters
Proceedings of the 66th MIPT All-Russian Scientific Conference
6. D. Dorin, N. Kiselev, A. Grabovoy
Spatial and temporal methods of time series analysis
Proceedings of the 66th MIPT All-Russian Scientific Conference

POSTER SESSIONS

- December 10, 2025
Kandinsky 5.0 Image Post-Train: SFT & RLHF
Salute, Giga!
- October 21, 2024
Spatio-Temporal fMRI Analysis in Visual Stimuli Decoding: Linear Model Forecasting & Voxel Weighing
Neuroinformatics 2024

TALKS

- September 28, 2026
Subspace of Maximum Curvature for Loss Landscape Stabilization
16th International Conference on Intelligent Data Processing: Theory and Applications
- May 29, 2026

Adaptive Selection of the Gaussian Sampling Scale for Monte Carlo Estimation of Loss-Landscape Convergence

Ivannikov Memorial Workshop

○ April 4, 2026

A mean-square criterion for loss-landscape convergence based on the leading eigenspace of the Hessian

68th MIPT All-Russian Scientific Conference

○ December 10, 2025

Robust Convergence of Loss Landscapes through Distributional Averaging

2025 Ivannikov ISP RAS Open Conference

○ December 10, 2025

Kandinsky: image and video diffusion models fine-tuning for personalization

Salute, Giga!

○ November 20, 2025

Kandinsky: launching and fine-tuning image and video generation models

AI Journey 2025

○ September 25, 2025

Sufficient sample size and its connection to the loss landscape convergence

22nd All-Russian Conference with International Participation “Mathematical Methods of Pattern Recognition”

○ September 25, 2025

Decoding visual stimuli from multimodal brain signals

22nd All-Russian Conference with International Participation “Mathematical Methods of Pattern Recognition”

○ April 5, 2025

Loss landscape convergence as an indicator of sufficient sample size

67th MIPT All-Russian Scientific Conference

○ December 12, 2024

Hessian-based convergence analysis of convolutional network loss landscapes

2024 Ivannikov ISP RAS Open Conference

○ April 6, 2024

Determining a sufficient sample size from the posterior distribution of model parameters

66th MIPT All-Russian Scientific Conference

○ April 6, 2024

Spatial-temporal methods for time series analysis

66th MIPT All-Russian Scientific Conference

TEACHING

My First Scientific Paper

Teaching assistant

Moscow Institute of Physics and Technology

Sep 2025 – Present

○ Playlists: Spring 2026, Spring 2025.

Diffusion Models

Instructor

Standard Data × Sber University

Nov 2024 – Present

○ Implementation Details: DDIM, noise schedulers, guidance overexposure, noise offset, multiple text encoders, transformer-based diffusion models, MM-DiT.

○ Video Diffusion Models: Deforum, AnimateDiff, Wan.

Deep Learning

Course organizer, Instructor

Moscow Institute of Physics and Technology

Sep 2024 – Present

○ Optimization, Regularization.

○ Initialization, Normalization, CNN.

○ Generative Models: Autoregression, VAE, GAN, Diffusion, Flow Matching.

○ Acceleration: Quantization, Pruning, Distillation, KV-Cache, Flash Attention.

○ Playlists: Fall 2026, Fall 2025, Fall 2024.

PROJECTS

- **HippoTrainer: Gradient-Based Hyperparameter Optimization**
A Python library for gradient-based hyperparameter optimization, implementing cutting-edge algorithms that leverage automatic differentiation to efficiently tune hyperparameters.
- **Just Relax It: Discrete Variables Relaxation**
A cutting-edge Python library designed to streamline the optimization of discrete probability distributions in neural networks, offering a suite of advanced relaxation techniques compatible with PyTorch.
- **Models of epidemic spread, in particular COVID-19 as a model of stochastic chemical kinetics**
Various approaches to modeling the spread of epidemics, differential equations and Markov processes
- **Optimization methods for quadratic problems with large dimensionality**
Comparison of different methods of solving high-dimensional linear regression problems
- **Intelligent Presentation Generator**
Application for generating presentations based on text files using topic modeling

ACHIEVEMENTS

- **Fall 2026-2027:**
 - Russian Presidential Scholarship
- **Spring 2025-2026:**
 - Russian Presidential Scholarship
 - Sber Scholarship Program (Top-91 students in Russia)
 - K.V. Rudakov scientific academic scholarship for research activities in the field of applied mathematics
 - Increased State Academic Scholarship for 4 year bachelor and master students at MIPT
- **Fall 2025-2026:**
 - Russian Presidential Scholarship
 - Sber Scholarship Program (Top-91 students in Russia)
 - K.V. Rudakov scientific academic scholarship for research activities in the field of applied mathematics
 - Increased State Academic Scholarship for 4 year bachelor and master students at MIPT
- **Spring 2024-2025:**
 - K.V. Rudakov scientific academic scholarship for research activities in the field of applied mathematics
 - Personal scholarship for contributions to the development of numerical optimization methods
 - Increased State Academic Scholarship for 4 year bachelor and master students at MIPT
- **Fall 2024-2025:**
 - K.V. Rudakov scientific academic scholarship for research activities in the field of applied mathematics
 - Personal scholarship for contributions to the development of numerical optimization methods
 - Increased State Academic Scholarship for 4 year bachelor and master students at MIPT
- **Spring 2023-2024:**
 - Personal scholarship for contributions to the development of numerical optimization methods
- **Fall 2023-2024:**
 - K.V. Rudakov scientific academic scholarship for research activities in the field of applied mathematics
 - Personal scholarship for contributions to the development of numerical optimization methods
- **2020-2023:**
 - Abramov scholarship for 1-3 year bachelor students with the best grades at MIPT

CERTIFICATIONS

- **Statistics for data analysis**
Coursera, Issued Jan 2022, Credential ID KAQRGNCQJ8AH
- **Unsupervised learning**
Coursera, Issued Jan 2022, Credential ID 3CTYTEFT48FM
- **Supervised learning**
Coursera, Issued Jan 2022, Credential ID 2ZBSN8L7EAVV
- **Mathematics and Python**
Coursera, Issued Oct 2021, Credential ID CSTTGDM8RF2V

SKILLS

- **DL:** PyTorch, Huggingface, Accelerate, Multi-GPU training: DDP/FSDP/SP/TP, W&B, TensorBoard
- **ML:** NumPy, SciPy, Pandas, NetworkX, Scikit-learn, LightGBM, CatBoost
- **OS:** macOS, Linux, Windows
- **Misc.:** Git, Bash, $\text{\LaTeX}$
- **Soft Skills:** responsible, organized, critical thinker, flexible, communicative, team player, patient

LANGUAGES

- Russian (Native)
- English (Advanced)

INTERESTS

- Coffee, Wine, Jazz
- Classic Guitar
- Gym, Hiking
- Traveling